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Rakib Hossan

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Research Interests

Large Language Models: scientific reasoning, claim verification, claim revision

Agentic LLMs and test-time reinforcement learning

Multilingual and low-resource NLP (Bangla), speech and clinical AI

Factuality and hallucination mitigation in LLMs

Education

Apr 2023–Jan 2027 **B.Sc. Engg. in Computer Science & Engineering**, *Bangladesh University of Business and Technology (BUBT)*, Dhaka, Bangladesh
CGPA – **4.00/4.00** (9th semester running)
Dean's Award for Academic Achievement – Spring 2025, Fall 2025
Thesis: claim verification and abductive reasoning with LLMs (in progress)

2012–2014 **Pre-Sea Training in Marine Engineering**, *United Marine Academy*, Dhaka, Bangladesh
CGPA – 3.66

2012 **Higher Secondary Certificate (Science)**, *Dhaka College*
GPA – 5.00/5.00

2010 **Secondary School Certificate (Science)**, *Dhanmondi Govt. Boys High School*
GPA – 5.00/5.00

Publications

BLP @ AACL 2025 [1] **Rakib Hossan**, and Shubhashis Roy Dipta. "PromptGuard at BLP-2025 Task 1: A Few-Shot Classification Framework Using Majority Voting and Keyword Similarity for Bengali Hate Speech Detection." Proceedings of the BLP Workshop @ AACL-IJCNLP 2025. [PDF] [Code]

Research Projects

Ongoing **Claim Revision Benchmark (Thesis)**
With Shubhashis Roy Dipta, Mir Sazzat Hossain, MD Fahim
- Building a benchmark that tests whether LLMs can revise a paper's claims when one result is withdrawn or corrected, using claim dependency graphs built from arXiv papers.
- Three tasks: withdrawal, numeric correction, and cross-paper propagation of real published corrections; includes trap claims that look related but do not depend on the changed result. Manuscript in preparation (ACL).

Jul 2026–Ongoing **Agentic Test-Time Reinforcement Learning**, *Penta AI team*
With MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Applying test-time reinforcement learning (TTRL) to LLM agents, so they can keep improving on new, unlabeled tasks using their own consensus as a reward signal.

Jul 2026–Ongoing **Bangla Dialectal Medical ASR Benchmark**, *Penta AI team*
With MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- A benchmark for clinically significant errors in Bangla dialectal medical speech recognition, built from synthetic doctor–patient conversations generated with LLMs and text-to-speech.

Jul 2026–Ongoing **Medical Prescription Understanding**, *Penta AI team*
With MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Reading and structuring medical prescriptions with AI, so that drug names, dosages, and instructions can be extracted reliably.

Industrial Experience

Apr 2026–Present **AI Researcher & Engineer**, *Penta Global Ltd.*, Dhaka
- Research on Bangla medical speech recognition, medical prescription understanding, and agentic test-time RL (main focus since Jul 2026).
- Designed LLM pipelines for claim extraction, evidence grounding, and dependency-graph validation over scientific documents.
- Applied RAG, **majority-vote consensus** across repeated generations, and **LLM-judge** quality scoring to reduce hallucination.

Aug 2026–Nov 2026 **AI Engineer**, *AVNBD*, Dhaka
- Built a multi-stage LLM pipeline to extract, structure, and translate a large-scale human-rights dataset from a scanned Bangla source.
- Cross-referenced extracted records against an independent third-party dataset to validate accuracy.

Jun 2025–Feb 2026 **Software Engineer**, *OpenCare*, Hybrid
- Institution management, Keycloak auth, MinIO file storage, Liquibase migrations, Dockerized deployment. [Contributors]

Apr 2024–Mar 2026 **Software Engineer**, *Penta Global Ltd.*, Dhaka
- Enterprise applications: Bangladesh Election Commission web system, ISP billing services, and the HRM Management System.

Honors & Awards

Jan 2026 **Dean's Award for Academic Achievement**, Faculty of Engineering & Applied Sciences, BUBT (Fall 2025)

Aug 2025 **Dean's Award for Academic Achievement**, Faculty of Engineering & Applied Sciences, BUBT (Spring 2025)

Skills

LLM Research	LLMs, RAG, prompt engineering, few-shot learning, abductive reasoning, LLM-as-judge evaluation
ML & NLP	Transformers, CNNs, RNNs/LSTMs, text classification, word embeddings
Frameworks	PyTorch, Hugging Face Transformers, TensorFlow, Keras, scikit-learn, Pandas, NLTK
Languages	Python, Java (Spring Boot), TypeScript (NestJS), C/C++
Systems	PostgreSQL, Redis, RabbitMQ, Keycloak, Docker, CI/CD

Certifications

Apr 2026 Natural Language Processing Specialization, DeepLearning.AI

Mar 2026 Deep Learning Specialization, DeepLearning.AI

Feb 2026 Machine Learning Specialization, DeepLearning.AI

Prior Career: Marine Engineering

Five years at sea on super tankers before computer science: safety-critical, team-coordinated work where every fault is traced to a root cause.

Mar 2022–Jan 2023 **Junior Engineer**, *MT. XIDI, Seacon Star*, At sea
- VLCC (GRT 160,000). Maintained the main engine, generator (Yanmar), fresh water generator, boiler (MHI/MAC-90 B), and purifiers.

Sep 2017–Aug 2021 **Engine Cadet**, *ATLANTIC 1 (Jhampur Group), MT. ARTAVIL (NITC)*, At sea
- Maintained the main engine (incl. Mitsubishi 7UEC85LS), generators, fresh water generator, S/W treatment plant, purifiers, and emergency machinery.

References

Available on request.